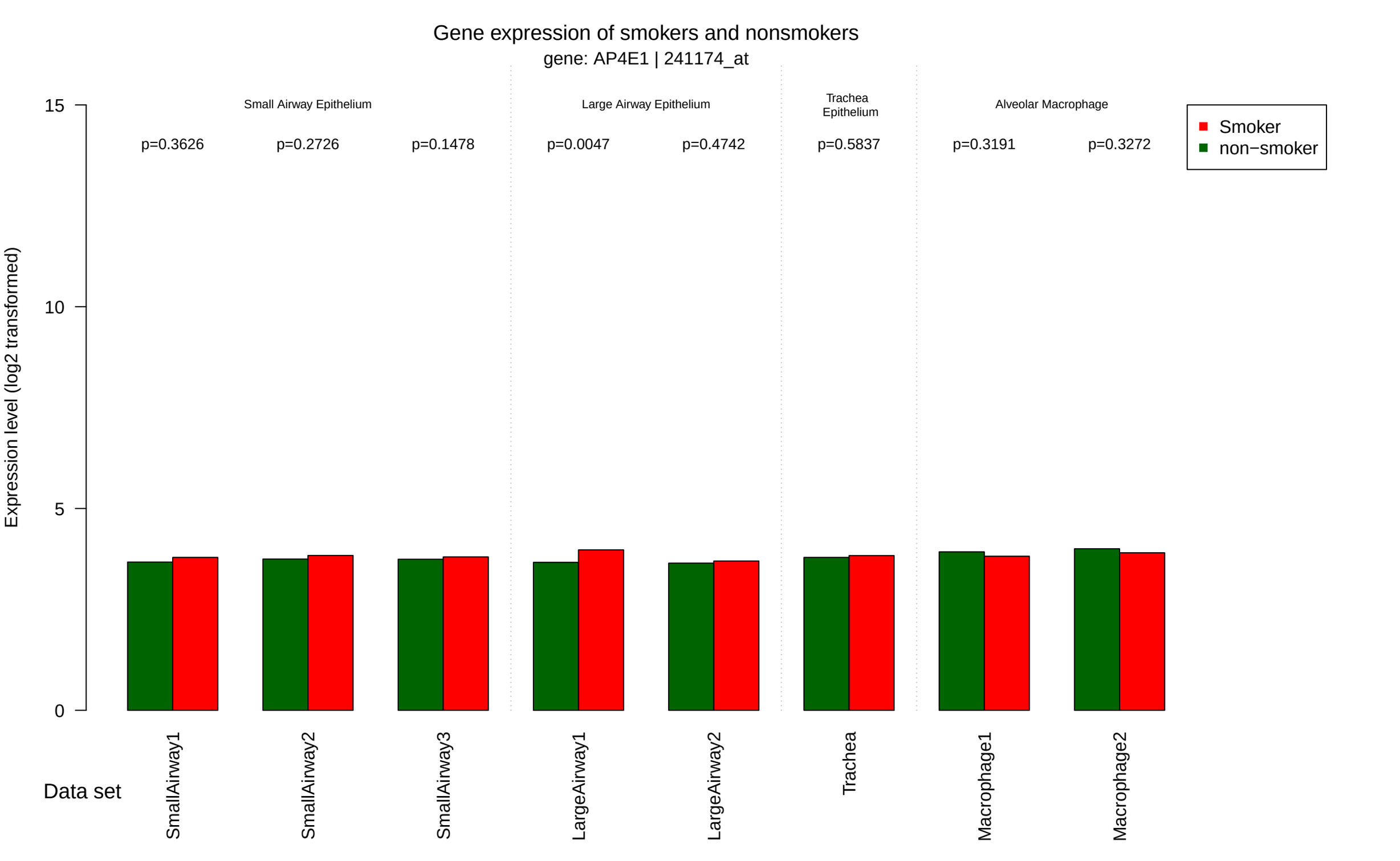
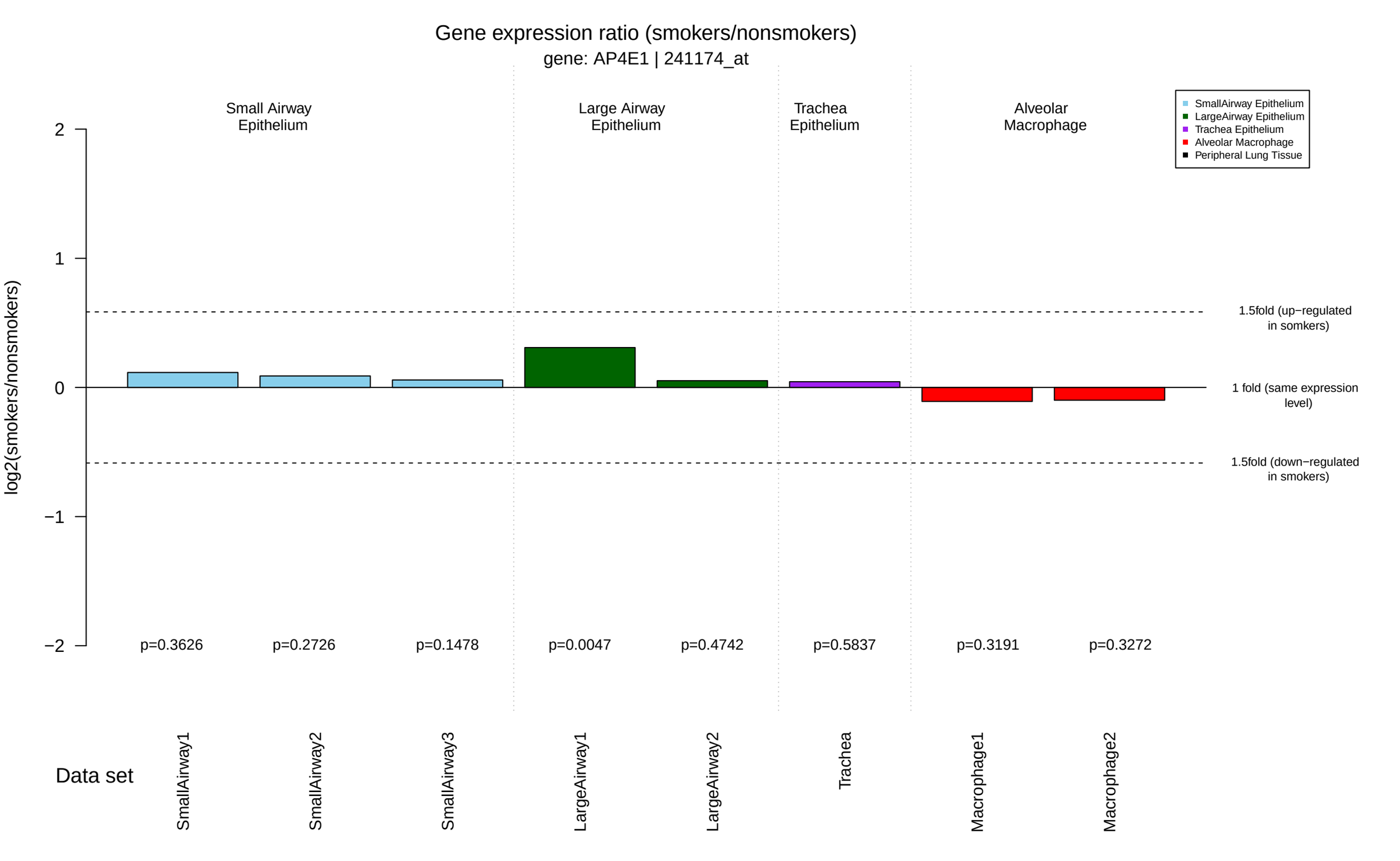
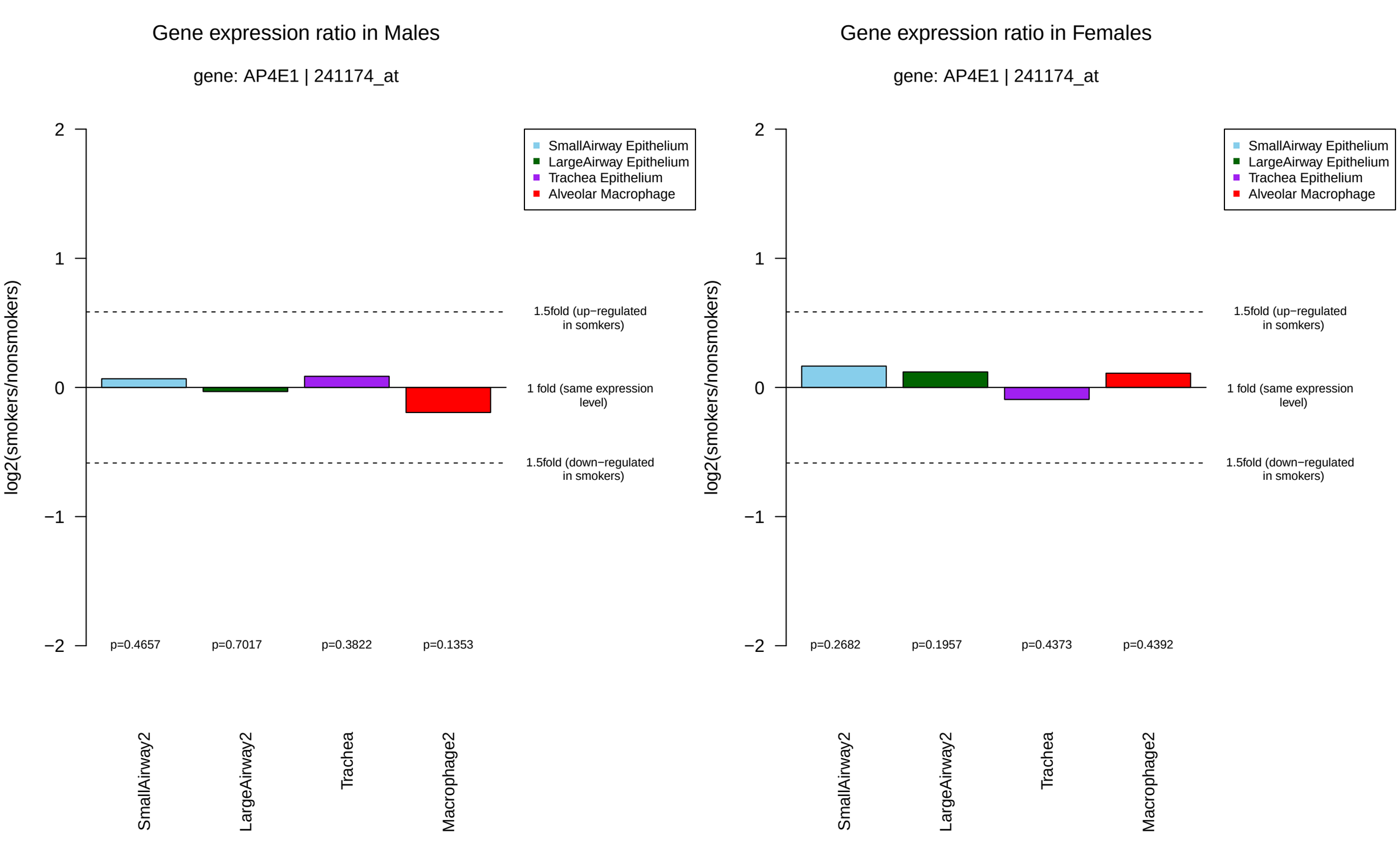
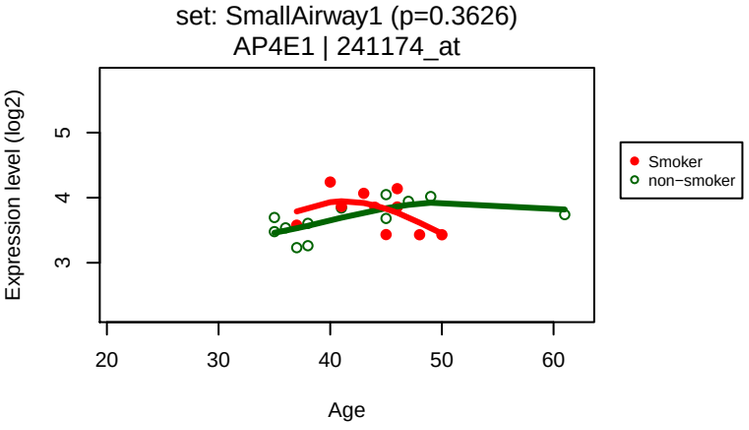
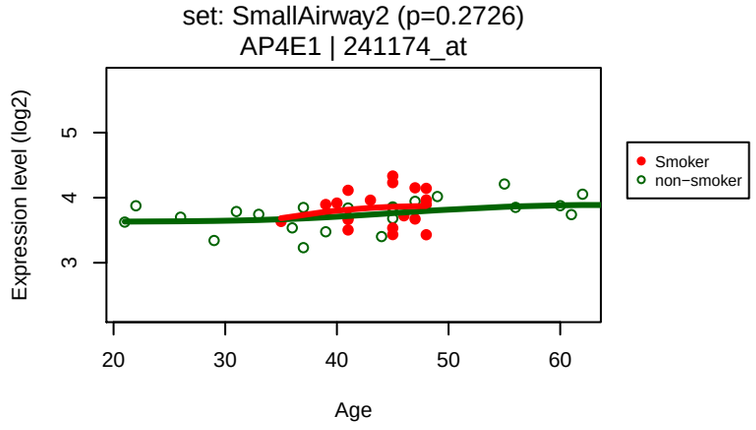
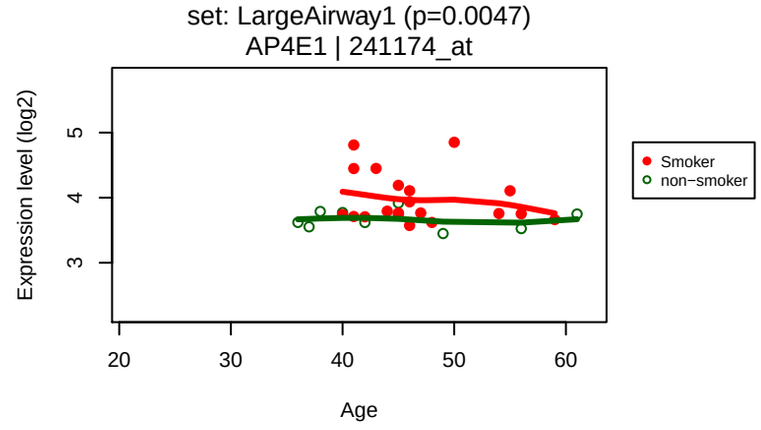

Gene expression of smokers and nonsmokers
gene: AP4E1 | 241174_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3626
p=0.2726
p=0.1478
p=0.0047
p=0.4742
p=0.5837
p=0.3191
p=0.3272
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

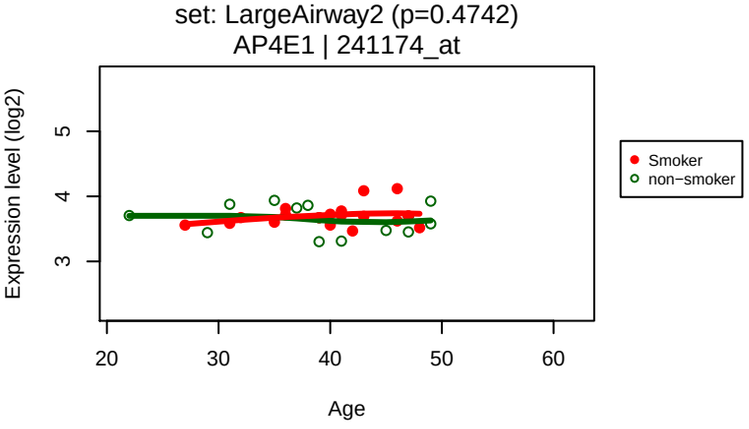

Gene expression ratio (smokers/nonsmokers)
gene: AP4E1 | 241174_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3626
p=0.2726
p=0.1478
p=0.0047
p=0.4742
p=0.5837
p=0.3191
p=0.3272
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

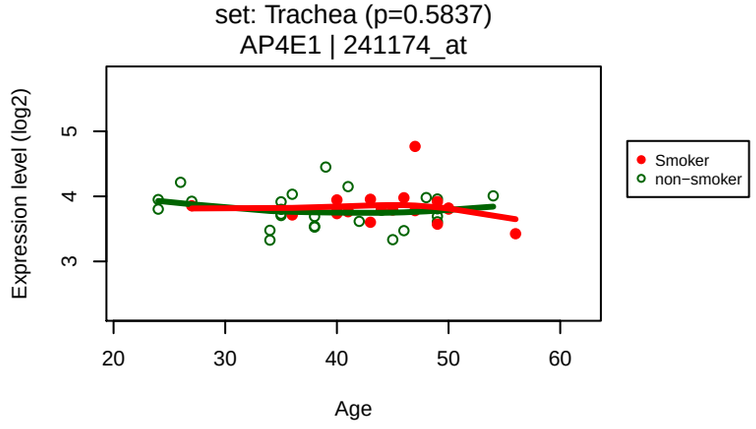

Gene expression ratio in Males
gene: AP4E1 | 241174_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4657
p=0.7017
p=0.3822
p=0.1353
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AP4E1 | 241174_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2682
p=0.1957
p=0.4373
p=0.4392
SmallAirway2
LargeAirway2
Trachea
Macrophage2

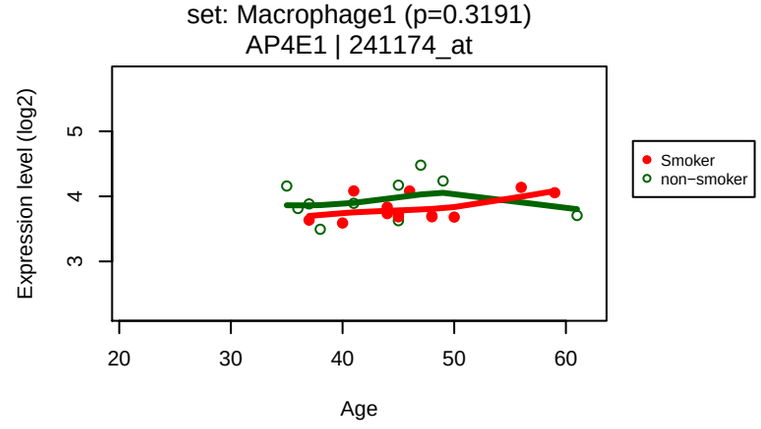

set: SmallAirway1 (p=0.3626)
AP4E1 | 241174_at
Expression level (log2)
Age
Smoker
non−smoker

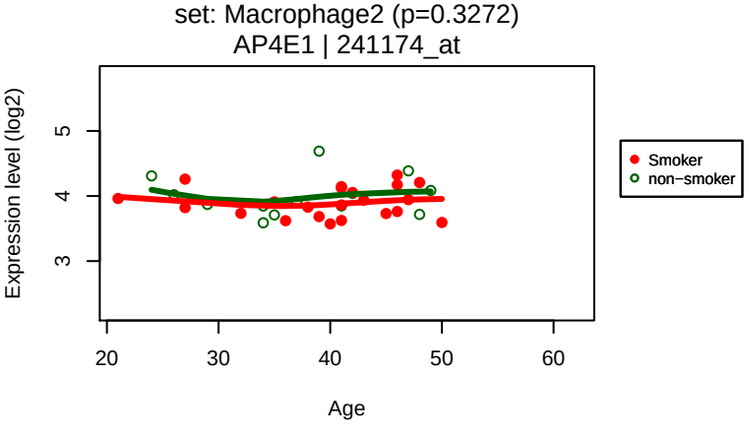

set: SmallAirway2 (p=0.2726)
AP4E1 | 241174_at
Expression level (log2)
Age
Smoker
non−smoker

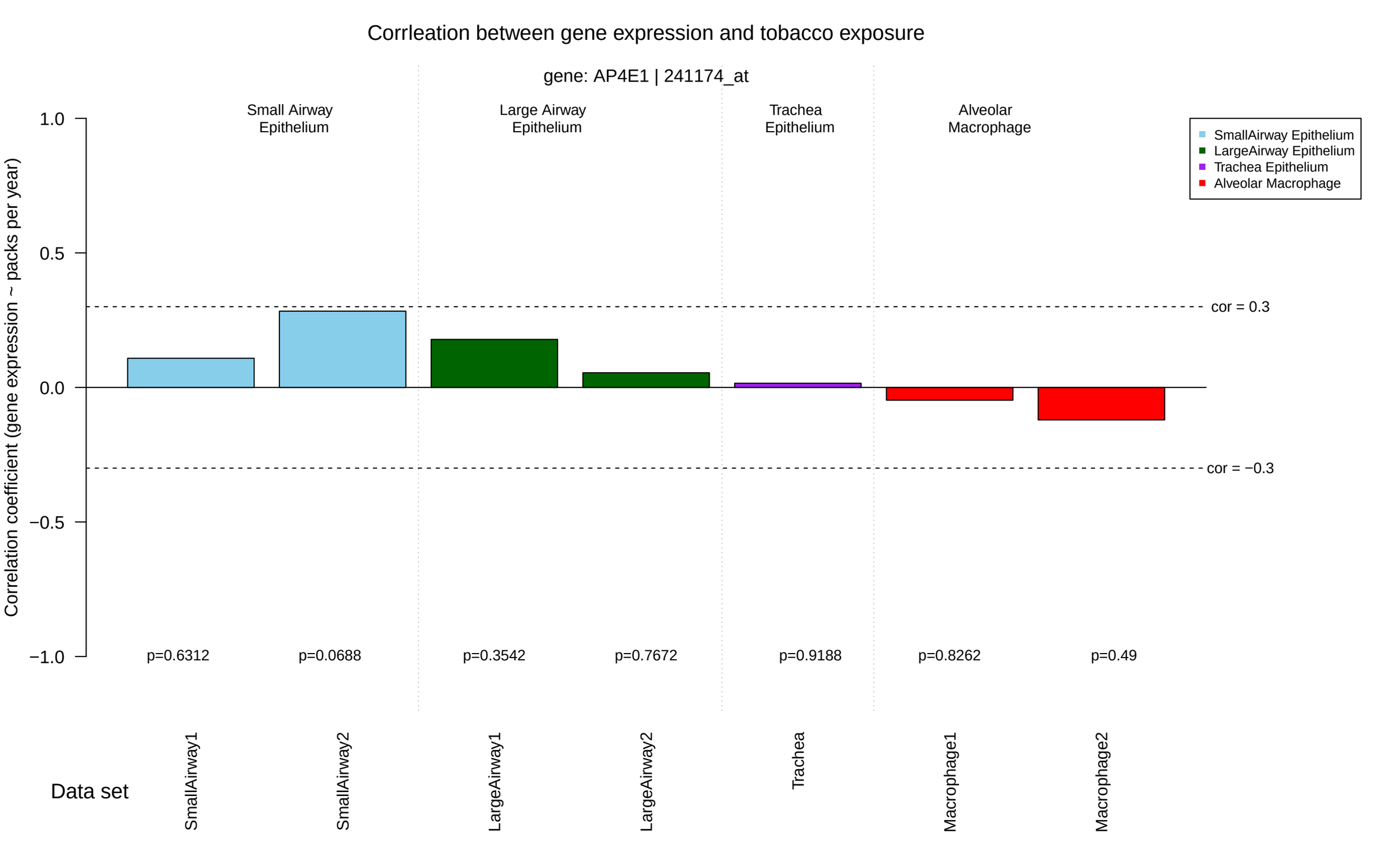

set: LargeAirway1 (p=0.0047)
AP4E1 | 241174_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4742)
AP4E1 | 241174_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5837)
AP4E1 | 241174_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3191)
AP4E1 | 241174_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3272)
AP4E1 | 241174_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AP4E1 | 241174_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6312
p=0.0688
p=0.3542
p=0.7672
p=0.9188
p=0.8262
p=0.49
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2